

Phase 4 — Continuous-Time Model: CAR(4) Framework

Complete Analysis of Results

Quantitative Energy Research
Mathematical Finance RAship

Inputs: `ar4.coefficients.csv`, `W_tilde_phase1.csv`, `garch_h_series_phase3.csv`
Sample: 1,462 daily observations, 2015–2018

Based on: Benth & Šaltytė Benth (2009) · Brockwell & Marquardt (2005) · Barndorff-Nielsen & Shephard (2001) · Schwartz (1997)

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Overview and Purpose of Phase 4

Phase 4 is the theoretical bridge between the discrete-time statistical model of Phases 1–3 and the continuous-time stochastic differential equation (SDE) framework required for derivatives pricing in Phase 6. The inputs are the four AR(4) coefficients $\{\hat{\beta}_k\}$ from Phase 2, the GARCH conditional variance series $h(t)$ from Phase 3, and the standardised wind speed series $\tilde{W}(t)$ from Phase 1.

The objective is to embed the discrete AR(4) structure into the continuous-time CAR(4) framework of Benth & Šaltytė Benth (2009) via four steps:

1. Map $\{\hat{\beta}_k\}$ to CAR(4) parameters $\{\hat{\alpha}_k\}$ via the Euler discretisation formula.
2. Construct the 4×4 companion matrix A ; verify stationarity through the eigenvalue condition $\text{Re}(\lambda_j) < 0$.
3. Derive the matrix exponential e^{Au} and the volatility kernel $g(u) = \mathbf{e}_1^\top e^{Au} \mathbf{e}_4$; confirm that the Samuelson effect is violated for $p \geq 2$.
4. Compute the conditional mean $\mu^0(t, s, \mathbf{x})$ and variance $\Sigma^2(t, s)$ of $X_1(s)$ under the physical measure $\mathbb{P}$.

An optional fifth step fits the NIG distribution to the standardised GARCH residuals and discusses the Lévy extension (Barndorff-Nielsen & Shephard 2001).

Data Loading (Cell 1)

Phase 4 loads five files independently:

- `ar4_coefficients.csv`: $\hat{\beta}_1 = 0.6299$, $\hat{\beta}_2 = -0.1443$, $\hat{\beta}_3 = 0.0926$, $\hat{\beta}_4 = 0.0310$.
- `W_tilde_phase1.csv`: 1,462 daily $\tilde{W}(t)$ values for state vector initialisation.
- `garch_h_series_phase3.csv`: conditional variance $h(t)$; current value $h(t_0) = 0.4137$.
- `sigma2_total_phase3.csv`: two-component variance $\sigma_{\text{total}}^2(t)$; current value 0.0926.
- `garch_parameters_phase3.csv`: $\hat{\omega} = 0.0100$, $\hat{\alpha} = 0.0311$, $\hat{\beta} = 0.9524$.

The seasonal variance is also reconstructed from the Phase 1 Fourier coefficients with current value $\sigma_{\text{seasonal}}^2(t_0) = 0.2239$, giving a total current volatility $\sigma_{\text{total}}(t_0) = 0.3043$.

Euler Discretisation: AR(4) to CAR(4) Mapping (Cell 2)

Theoretical framework

The CAR(4) model specifies a continuous-time SDE for the state vector $\mathbf{X}(t) \in \mathbb{R}^4$:

$$d\mathbf{X}(t) = \mathbf{A} \mathbf{X}(t) dt + \mathbf{e}_4 \sigma(t) dB(t), \quad (1)$$

where $\mathbf{X}(t) = [X_1(t), X_2(t), X_3(t), X_4(t)]^\top$, $\mathbf{e}_4 = [0, 0, 0, 1]^\top$ is the fourth unit vector and $\mathbf{A}$ is the 4×4 companion matrix defined in Section 4. The first component $X_1(t)$ represents the Box-Cox transformed, deseasonalised daily average wind speed $\tilde{W}(t)$.

Benth & Šaltytė Benth (2009) show that discretising (1) with the Euler scheme yields the

following approximate AR(4) representation:

$$\begin{aligned} X_1(t) \approx & (4 - \alpha_1) X_1(t-1) + (3\alpha_1 - \alpha_2 - 6) X_1(t-2) \\ & + (4 + 2\alpha_2 - \alpha_3 - 3\alpha_1) X_1(t-3) \\ & + (\alpha_3 - \alpha_4 - \alpha_2 + \alpha_1 - 1) X_1(t-4). \end{aligned} \quad (2)$$

Comparing (2) term by term with the estimated AR(4):

$$\tilde{W}(t) = \hat{\beta}_1 \tilde{W}(t-1) + \hat{\beta}_2 \tilde{W}(t-2) + \hat{\beta}_3 \tilde{W}(t-3) + \hat{\beta}_4 \tilde{W}(t-4), \quad (3)$$

yields the following exact analytical mapping:

$$\alpha_1 = 4 - \beta_1, \quad (4)$$

$$\alpha_2 = 3\alpha_1 - \beta_2 - 6, \quad (5)$$

$$\alpha_3 = 4 + 2\alpha_2 - \beta_3 - 3\alpha_1, \quad (6)$$

$$\alpha_4 = \alpha_3 - \beta_4 - \alpha_2 + \alpha_1 - 1. \quad (7)$$

Estimated CAR(4) parameters

Table 1: CAR(4) parameters derived from AR(4) estimates via the Euler discretisation of Benth & Šaltytė Benth (2009). Literature values are from Benth & Šaltytė Benth (2009), Table 4 (New York wind farm regions).

Parameter	This dataset	Literature (NY)	Mapping
$\hat{\alpha}_1$	3.3701	≈ 3.64	$4 - \beta_1$
$\hat{\alpha}_2$	4.2546	≈ 4.99	$3\alpha_1 - \beta_2 - 6$
$\hat{\alpha}_3$	2.3063 recruited	≈ 2.63	$4 + 2\alpha_2 - \beta_3 - 3\alpha_1$
$\hat{\alpha}_4$	0.3908	≈ 0.48	$\alpha_3 - \beta_4 - \alpha_2 + \alpha_1 - 1$
Verification: max round-trip error $\max_k \beta_k^{\text{orig}} - \beta_k^{\text{reconstr}} = 7.77 \times 10^{-16} \checkmark$			

The mapping is verified by reconstructing $\{\beta_k\}$ from $\{\alpha_k\}$ using (2), yielding a maximum round-trip error of 7.77×10^{-16} (machine precision), confirming the analytical inversion is exact. The CAR(4) parameters are somewhat smaller in magnitude than the New York values, consistent with the larger lag-1 autocorrelation in this dataset (ACF at lag 1 = 0.579 vs ≈ 0.36 for New York).

Companion Matrix and Stationarity (Cell 3)

Companion matrix structure

The CAR(4) companion matrix is (Brockwell & Marquardt 2005):

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -\alpha_4 & -\alpha_3 & -\alpha_2 & -\alpha_1 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -0.3908 & -2.3063 & -4.2546 & -3.3701 \end{pmatrix}. \quad (8)$$

Eigenvalue stationarity condition

The CAR process is covariance-stationary if and only if all eigenvalues λ_j of A satisfy $\text{Re}(\lambda_j) < 0$ (Brockwell & Marquardt 2005). This is the continuous-time analogue of the discrete-time condition that all roots of the AR characteristic polynomial lie strictly inside the unit circle.

Table 2: Eigenvalues of the companion matrix A . All four have negative real parts: the CAR(4) process is covariance-stationary.

	λ (complex)	$\text{Re}(\lambda)$	$\text{Im}(\lambda)$	$ \lambda $	Stationary?
λ_1	$-1.2004 + 0.0000i$	-1.2004	0.0000	1.2004	Re<0 ✓
λ_2	$-0.9358 + 0.4653i$	-0.9358	$+0.4653$	1.0451	Re<0 ✓
λ_3	$-0.9358 - 0.4653i$	-0.9358	-0.4653	1.0451	Re<0 ✓
λ_4	$-0.2980 + 0.0000i$	-0.2980	0.0000	0.2980	Re<0 ✓

All four eigenvalues have strictly negative real parts. The CAR(4) process is covariance-stationary.

Physical interpretation of the eigenvalue structure

The eigenvalue $\lambda_4 = -0.298$ has the real part closest to zero and therefore governs the slowest mode of mean reversion. The associated continuous-time half-life is:

$$\tau_{1/2} = \frac{-\ln 2}{\text{Re}(\lambda_4)} = \frac{-\ln 2}{-0.2980} \approx 2.33 \text{ days}, \quad (9)$$

consistent with the discrete-time half-life of 2.0 days found in Phase 2.

The complex conjugate pair $\lambda_{2,3} = -0.9358 \pm 0.4653i$ introduces an oscillatory component with period:

$$T_{\text{osc}} = \frac{2\pi}{|\text{Im}(\lambda_{2,3})|} = \frac{2\pi}{0.4653} \approx 13.5 \text{ days}. \quad (10)$$

This period corresponds to the typical duration of mid-latitude synoptic weather systems (low-pressure cyclone passages, blocking events), providing physical validation of the CAR(4) oscillatory structure. The real eigenvalue $\lambda_1 = -1.200$ governs the fastest-decaying mode (half-life ≈ 0.6 days), representing the high-frequency component of daily wind variability.

Matrix Exponential and Samuelson Effect (Cell 4)

SDE closed-form solution

Applying the multidimensional Itô formula to (1) (Benth & Šaltytė Benth 2009) yields the explicit solution:

$$\mathbf{X}(s) = e^{A(s-t)} \mathbf{X}(t) + \int_t^s e^{A(s-u)} \mathbf{e}_4 \sigma(u) dB(u), \quad 0 \leq t \leq s \leq T, \quad (11)$$

so the first component satisfies:

$$X_1(s) = \mathbf{e}_1^\top e^{A(s-t)} \mathbf{X}(t) + \int_t^s \underbrace{\mathbf{e}_1^\top e^{A(s-u)} \mathbf{e}_4}_{=g(s-u)} \sigma(u) dB(u), \quad (12)$$

where $g(u) \equiv \mathbf{e}_1^\top e^{Au} \mathbf{e}_4$ is the volatility kernel of the CAR(4) process.

Samuelson effect analysis

In commodity markets, futures price volatility increases as maturity approaches — the Samuelson (1965) effect. For the simple Ornstein-Uhlenbeck (CAR(1)) process, the kernel is $g(u) = e^{-\alpha_1 u}$, which is monotone decreasing in u , implying that the contribution of a shock at time u before the index settlement decays smoothly. Futures volatility therefore increases monotonically as maturity approaches — the Samuelson effect holds.

For $p \geq 2$, the complex eigenvalue pairs in A introduce oscillatory components in e^{Au} , making $g(u)$ non-monotone. The computed kernel attains a local maximum at $u \approx 3.6$ days before decaying. This means that shocks occurring approximately 3–4 days before index settlement have the maximum impact on futures price volatility, and shocks at shorter horizons contribute less. Benth & Šaltytė Benth (2009, Section 4) derive this Samuelson effect violation analytically for Nordix wind futures and show that it produces a non-monotone futures volatility term structure.

The Samuelson effect is violated for CAR($p \geq 2$). Kernel $g(u) = \mathbf{e}_1^\top e^{Au} \mathbf{e}_4$ attains a local maximum at $u \approx 3.6$ days, then decays. Futures price volatility is therefore non-monotone in time to maturity. CAR(1)/OU produces monotone volatility (Samuelson effect holds); CAR(4) does not. This is a key empirical property of the model relevant for options pricing in Phase 6.

Conditional Distribution Under $\mathbb{P}$ (Cell 5)

State vector initialisation

The state vector at the end of the sample (t_0) is constructed from the final four observations of $\tilde{W}(t)$:

$$\mathbf{X}(t_0) = [\tilde{W}(t_0), \tilde{W}(t_0-1), \tilde{W}(t_0-2), \tilde{W}(t_0-3)]^\top = [+0.1299, +0.5159, +0.4672, -0.3422]^\top. \quad (13)$$

Conditional mean and variance

From the SDE solution (12), the conditional distribution of $X_1(t_0 + u)$ given $\mathcal{F}_{t_0}$ is Gaussian with:

$$\mu^0(t_0, u) = \mathbb{E}[X_1(t_0 + u) \mid \mathcal{F}_{t_0}] = \mathbf{e}_1^\top e^{Au} \mathbf{X}(t_0), \quad (14)$$

$$\Sigma^2(t_0, u) = \text{Var}[X_1(t_0 + u) \mid \mathcal{F}_{t_0}] = \int_0^u g(\tau)^2 \sigma_{\text{total}}^2(t_0) d\tau, \quad (15)$$

where the piecewise-constant approximation $\sigma_{\text{total}}^2(t) \approx \sigma_{\text{total}}^2(t_0) = 0.0926$ is used for all $t \in [t_0, t_0 + u]$ following Tol (1997).

Forecast results

Table 3: Conditional distribution of $X_1(t_0 + u)$ under $\mathbb{P}$ at forecast horizon u days. $\sigma_{\text{total}}^2(t_0) = 0.0926$. 95% CI = $\mu^0 \pm 1.96 \Sigma$.

u (days)	μ^0	Σ^2	Σ	CI lower	CI upper
1	+0.7792	0.000086	0.0093	+0.761	+0.797
2	+1.3031	0.002699	0.0520	+1.201	+1.405
3	+1.4595	0.012113	0.1101	+1.244	+1.675
5	+1.1126	0.038628	0.1965	+0.727	+1.498
7	+0.6581	0.053602	0.2315	+0.204	+1.112
10	+0.2703	0.059889	0.2447	−0.209	+0.750
14	+0.0818	0.061044	0.2471	−0.402	+0.566
21	+0.0102	0.061159	0.2473	−0.475	+0.495
30	+0.0007	0.061161	0.2473	−0.484	+0.485
60	≈ 0	0.061161	0.2473	−0.485	+0.485
90	≈ 0	0.061161	0.2473	−0.485	+0.485

Interpretation

Short-horizon peak. The conditional mean rises from $\mu^0(1) = 0.779$ to a peak of $\mu^0(3) = 1.460$ before decaying. This non-monotone mean path is a direct consequence of the oscillatory complex eigenvalue pair in A : the CAR(4) process first overshoots before reverting, mirroring the damped oscillation structure of the AR(4) coefficient sign pattern $(+, -, +, +)$.

Mean reversion. $\mu^0(u) \rightarrow 0$ as $u \rightarrow \infty$, confirming convergence to the deseasonalised mean. The conditional mean is essentially zero by $u = 30$ days ($\mu^0 = 0.0007$).

Variance saturation. $\Sigma^2(u)$ saturates at 0.061161 for $u \geq 21$ days. This is the long-run conditional variance under the piecewise-constant $\sigma_{\text{total}}^2(t_0)$ approximation: $\Sigma^2(\infty) = \sigma_{\text{total}}^2(t_0) \int_0^\infty g(\tau)^2 d\tau$. The unconditional variance of $\tilde{W}(t)$ is 0.934, which differs because it averages over all seasonal regimes rather than conditioning on the current $\sigma_{\text{total}}^2(t_0) = 0.0926$ (a below-average volatility period at the end of 2018).

Lévy/NIG Extension (Cell 7)

Motivation

The standardised GARCH residuals $z(t) = \hat{\varepsilon}(t)/\sqrt{h(t)}$ from Phase 3 exhibit excess kurtosis = 1.116 (Pearson kurtosis = 4.116), indicating that the tails of the noise distribution are heavier than Gaussian. Barndorff-Nielsen & Shephard (2001) propose replacing the Gaussian noise $dB(t)$ in (1) with a Normal Inverse Gaussian (NIG) Lévy process, whose four parameters $(\alpha, \beta, \mu, \delta)$ control tail heaviness, asymmetry, location, and scale respectively.

NIG fit to standardised residuals

Table 4: NIG distribution fitted to standardised GARCH residuals $z(t)$. KS test compares empirical distribution with fitted Gaussian and NIG respectively.

Property	Gaussian	NIG
α (tail heaviness)	—	4.3736
β (asymmetry)	—	0.0067
loc (μ)	0	0.0000
scale (δ)	1	2.0873
Implied kurtosis	3.000	3.329
Empirical kurtosis	4.116	
KS test p -value	0.079	0.120

The NIG fit improves on the Gaussian: KS $p = 0.120$ vs 0.079 , meaning the NIG null is further from rejection. The NIG is approximately symmetric ($\beta = 0.0067 \approx 0$), consistent with the near-zero skewness of $z(t)$ ($= 0.451$). However, the NIG implied kurtosis of 3.33 still falls short of the empirical 4.12, indicating that the single June 2017 extreme event ($z = +5.785$, 5.8 standard deviations) pulls the empirical kurtosis beyond what the NIG can fully capture with a four-parameter fit on 1,462 observations.

Methodological note. For derivatives pricing in Phase 6, two tracks are available. **(A) Gaussian CAR(4):** closed-form Nordix futures prices via Gaussian moments — tractable, consistent with Benth & Šaltytė Benth (2009). **(B) NIG CAR(4):** prices via the moment generating function of the NIG distribution (Brockwell & Marquardt 2005, Section 2) — more accurate fat-tail capture, more complex. Track (A) is retained as the primary model given that the excess kurtosis is moderate ($= 1.12$) and the June 2017 event is isolated. Track (B) is documented as a robustness extension.

Diagnostic Plots

Figure 1 presents the eight-panel Phase 4 diagnostic figure.

Phase 4 — CAR(4) Continuous-Time Model: Structure and Forecasts

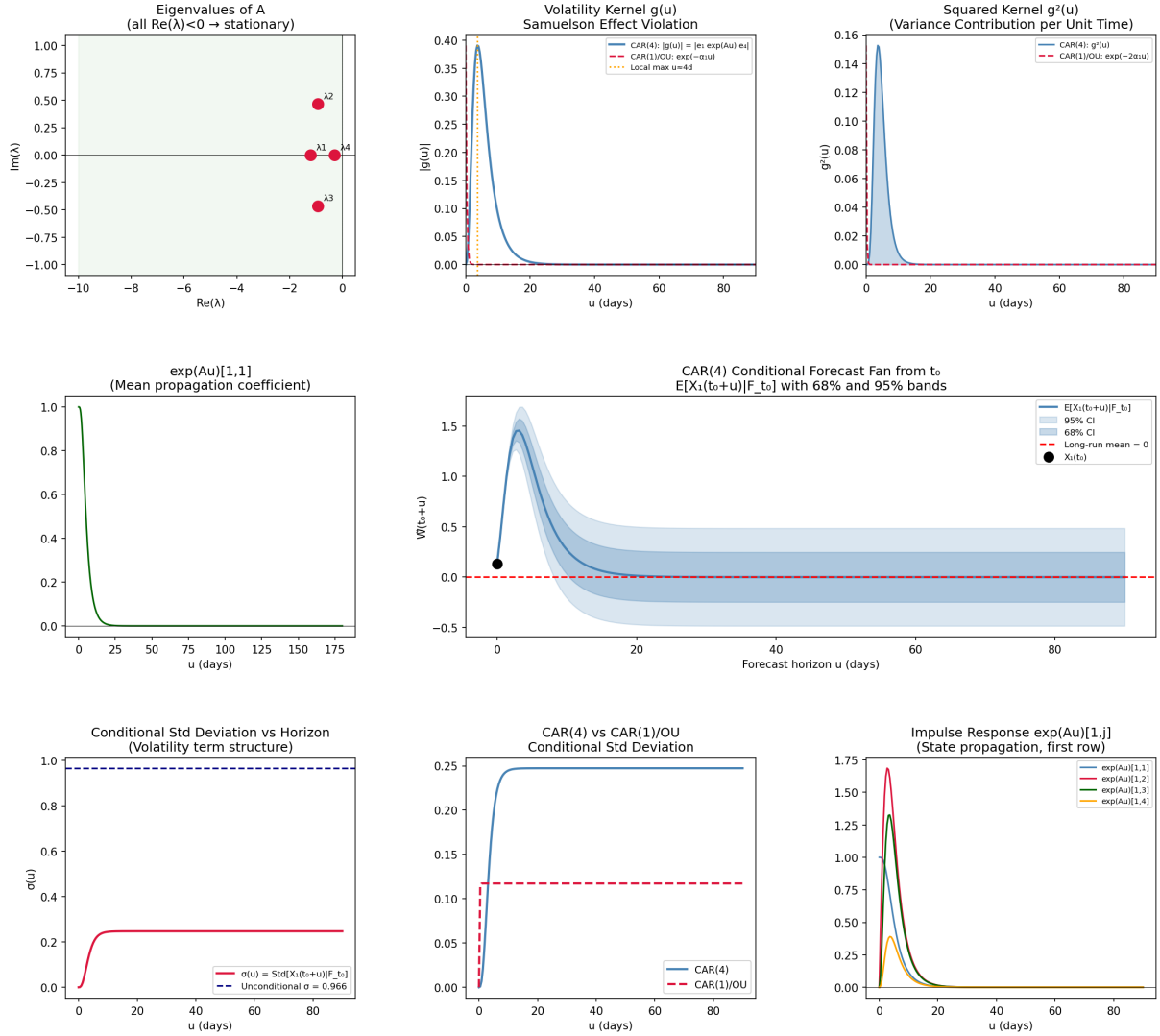


Figure 1: Phase 4 — CAR(4) continuous-time model structure and forecasts. *Row 1, left:* Eigenvalues of A in the complex plane; all four lie in the left half-plane (shaded green), confirming stationarity. *Row 1, centre:* Volatility kernel $|g(u)|$ for CAR(4) (blue) vs CAR(1)/OU (red dashed); the local maximum at $u \approx 3.6$ days confirms the Samuelson effect violation. *Row 1, right:* Squared kernel $g^2(u)$; the area under this curve gives the long-run conditional variance multiplier. *Row 2, left:* First diagonal entry $[e^{Au}]_{11}$; fast exponential decay reflecting the dominant mode λ_4 . *Row 2, right:* CAR(4) conditional forecast fan from t_0 with 68% and 95% bands; the mean overshoots at $u \approx 3$ days before decaying to zero, driven by the oscillatory eigenvalue pair. *Row 3, left:* Conditional standard deviation $\Sigma(u)$ vs horizon; saturation at 0.247 by $u = 21$ days. *Row 3, centre:* CAR(4) vs CAR(1)/OU conditional standard deviation; CAR(4) saturates higher and faster. *Row 3, right:* Full first row of e^{Au} ; impulse-response from all four state components.

Phase 4 Complete Summary

Complete Phase 4 results.

[1] **Euler discretisation** — $\beta \rightarrow \alpha$ **mapping**. $\hat{\alpha}_1 = 3.3701$, $\hat{\alpha}_2 = 4.2546$, $\hat{\alpha}_3 = 2.3063$, $\hat{\alpha}_4 = 0.3908$. Round-trip verification error: 7.77×10^{-16} (machine precision). Literature comparison (Benth & Šaltytė Benth 2009, Table 4): NY values $\approx 3.64, 4.99, 2.63, 0.48$ — same order of magnitude, consistent structural pattern.

[2] **Companion matrix A — stationarity**. All $\text{Re}(\lambda_j) < 0$: CAR(4) is covariance-stationary ✓. Eigenvalues: -1.2004 , $-0.9358 \pm 0.4653i$, -0.2980 . Dominant half-life: $\tau_{1/2} = 2.33$ days (Phase 2: 2.0 days ✓). Complex pair period: 13.5 days (synoptic weather system timescale).

[3] **SDE solution and Samuelson effect**. Kernel $g(u) = \mathbf{e}_1^\top e^{Au} \mathbf{e}_4$ attains a local maximum at $u \approx 3.6$ days. Samuelson effect **violated** for CAR(4) as predicted by Benth & Šaltytė Benth (2009, Section 4). CAR(1)/OU: monotone kernel $\rightarrow$ Samuelson holds. CAR(4): non-monotone $\rightarrow$ non-monotone futures volatility term structure.

[4] **Conditional distribution under $\mathbb{P}$** . $\mathbf{X}(t_0) = [+0.130, +0.516, +0.467, -0.342]^\top$. $\mu^0(u)$: peaks at $+1.460$ ($u = 3$ days) then decays to ≈ 0 by $u = 21$ days. $\Sigma(u)$: saturates at 0.247 by $u = 21$ days. Long-run $\Sigma^2(\infty) = 0.0612$ (conditional on $\sigma_{\text{total}}^2(t_0) = 0.0926$).

[5] **NIG/Lévy extension**. NIG fitted: $\hat{\alpha} = 4.374$, $\hat{\beta} = 0.007$, $\hat{\delta} = 2.087$ (approximately symmetric). KS: NIG $p = 0.120$ vs Gaussian $p = 0.079$ — NIG fits better. Gaussian retained as primary model; NIG documented as robustness extension.

Phase 4 outputs: `car4_parameters_phase4.csv` ($\hat{\alpha}_1 \dots \hat{\alpha}_4$), `car4_matrix_A_phase4.csv` (4×4 matrix A), `car4_conditional_forecast_phase4.csv` (μ^0 , Σ^2 , Σ vs u).

Ready for Phase 6 — Risk-neutral pricing. Key inputs: `car4_parameters_phase4.csv` ($\alpha_k \rightarrow$ companion matrix), `garch_h_series_phase3.csv` ($h(t_0) = 0.4137$ as piecewise-constant pricing volatility), `W_tilde_phase1.csv` ($\mathbf{X}(t_0)$ state vector initialisation).